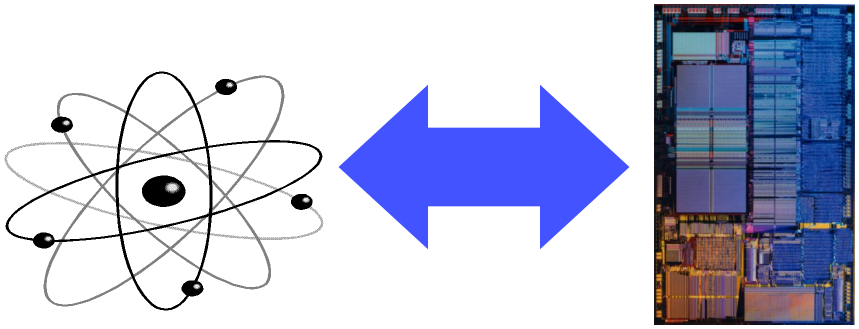
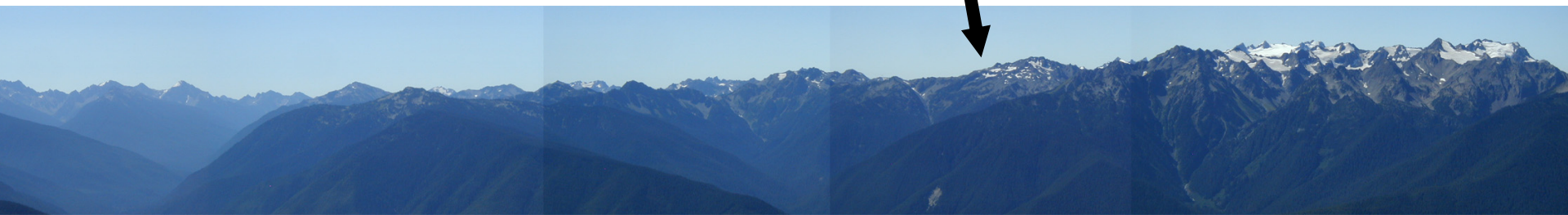
Architectures for Quantum Computers

CSE 471 Lecture
June 2, 2006

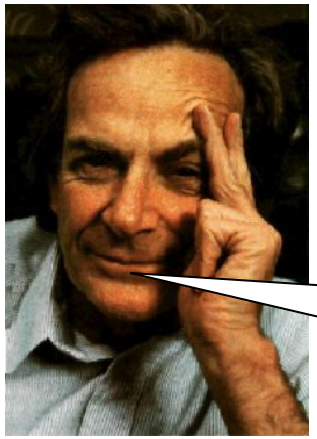
Dave Bacon



Soon **summer** will be here and you can be over here!



Quantum Slander



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Anyone who is not shocked by quantum theory has not understood it.



Niels Bohr
Nobel Prize 1922



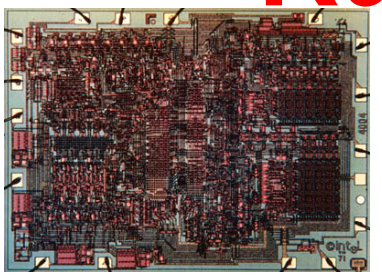
Do not be intemidated by quantum physicists!

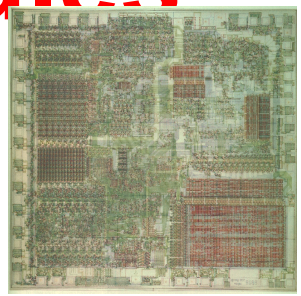
Today: Into the Quantum Void

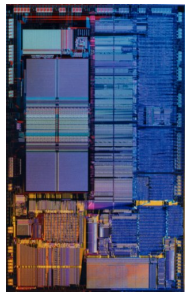
- What is a quantum computer?
- What, **really**, is a quantum computer?
- DiVincenzo's criteria
- Architectures for quantum computers

(What really really will a quantum computer look like?)

New Sizes, New Rules



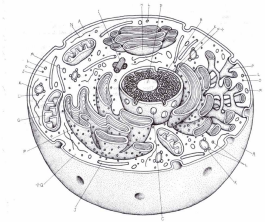




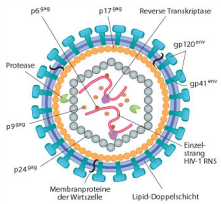
NOISY (FAULTY)

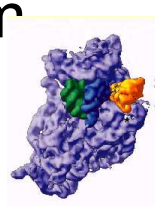
QUANTUM

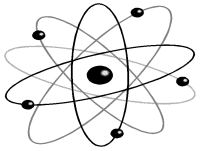
10,000nm 1,000nm 100nm 10nm 1nm 0.1nm











1. Can we build digital devices?
2. Quantum computers?

Why would we want to build here?

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

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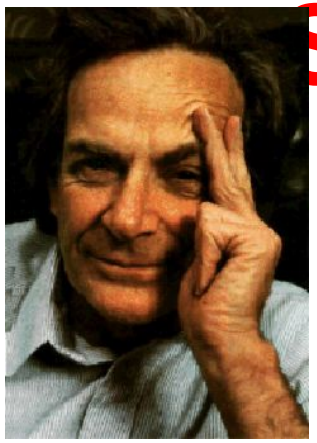


Peter Shor

1994

- Quantum computers can efficiently factor whole numbers (and compute discrete log.)
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Quantum Simulations

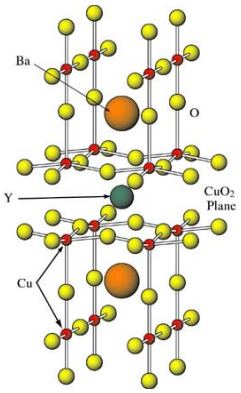


Richard Feynman

Simulating quantum systems on classical computers is hard

Simulating quantum systems on quantum computers is easy

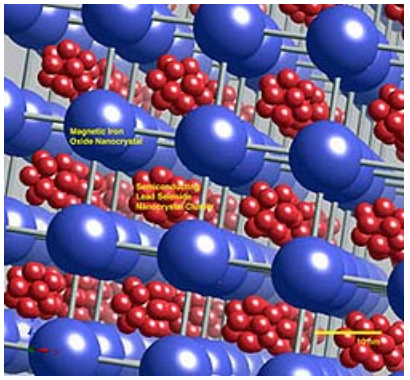
Quantum algorithms for simulating quantum processes



high temperature superconductors



biochemistry



novel materials

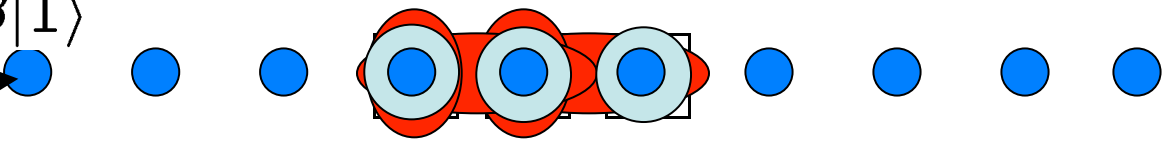
Bit, Pbit, and Qubit

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent 1 *1-p* percent	0 $\lvert\alpha\rvert^2$ percent 1 $\lvert\beta\rvert^2$ percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic	unitary

What is a Quantum Computer?

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

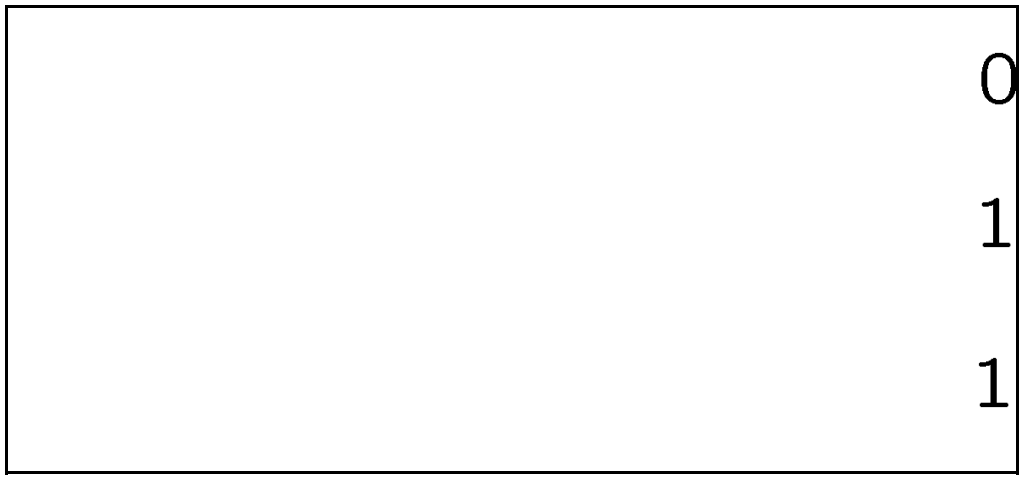
qubits



$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

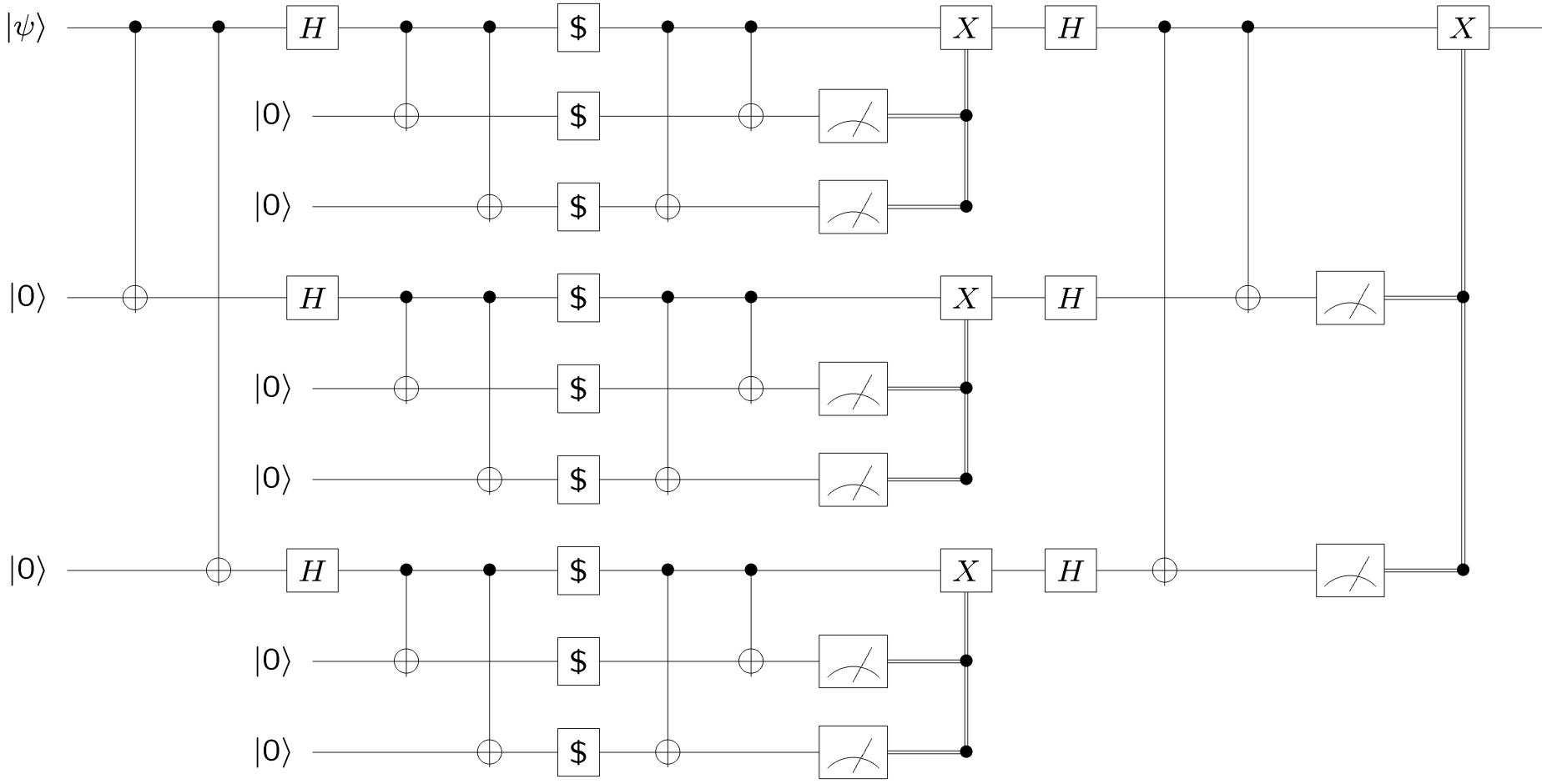
Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement



Quantum Circuits

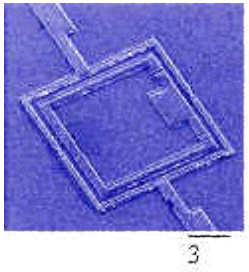
Quantum algorithms are usually expressed as families of quantum circuits



WARNING: quantum computers will (probably) not be made of etched circuit boards corresponding to the above circuits. Quantum circuits = Quantum algorithms

But What Will It Look Like?

Solid State



superconducting circuits

electron spin in Phosphorus doped Silicon

quantum dots

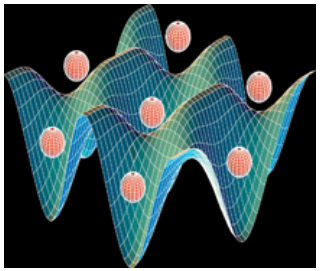
defects in diamonds

Atomic

cavity QED

neutral atoms in optical lattices





Molecular

Liquid NMR (no longer?)

Photon Based

linear optics plus single photon devices

Pics: Mabuchi (Caltech), Orlando (MIT)

DiVincenzo's Criteria



David DiVincenzo

1. Well defined qubits in a "scalable" architecture

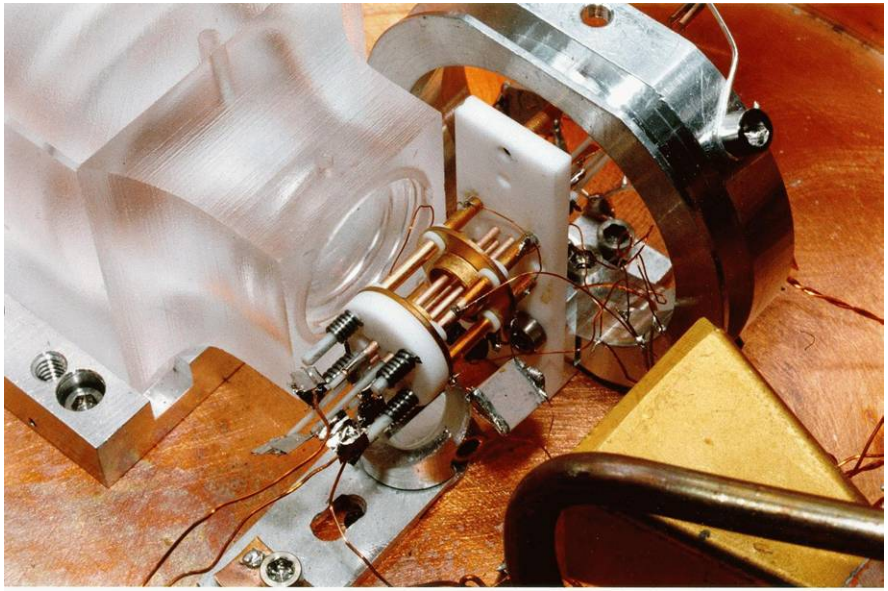
2. The ability to initialize the system to a state

3. Have faster control over the system than error processes in the system.

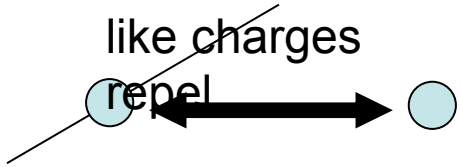
4. Have the ability to perform a universal set of quantum gates.

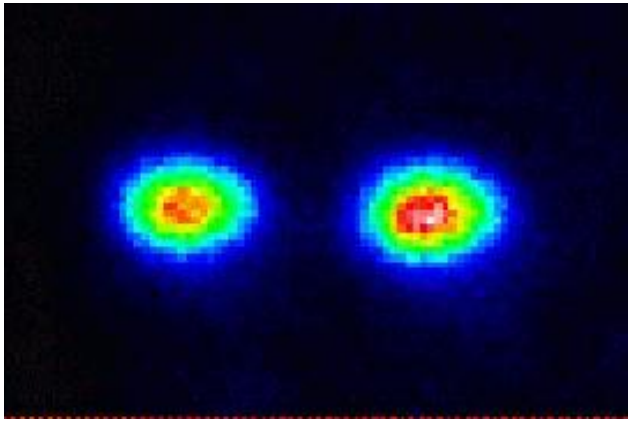
5. Have the ability to perform high quality measurements

Ion Trap

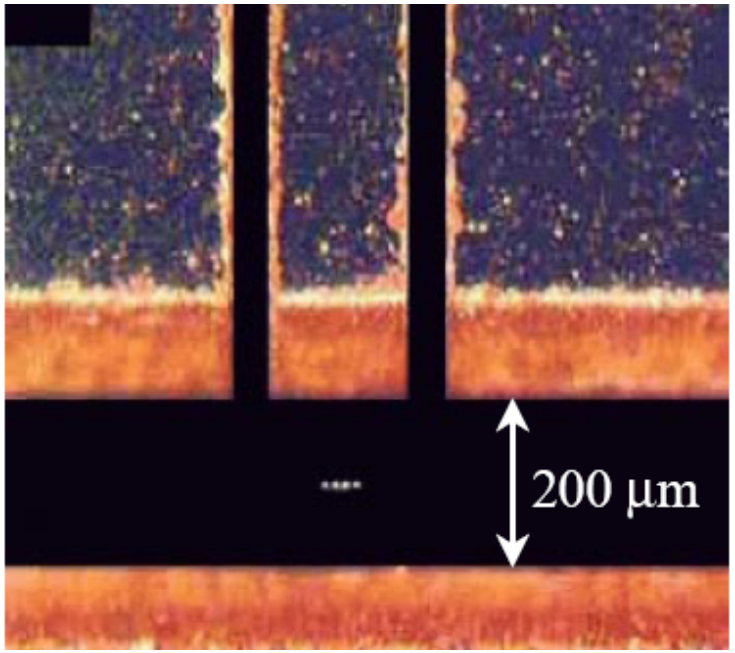


Oscillating electric fields trap ions

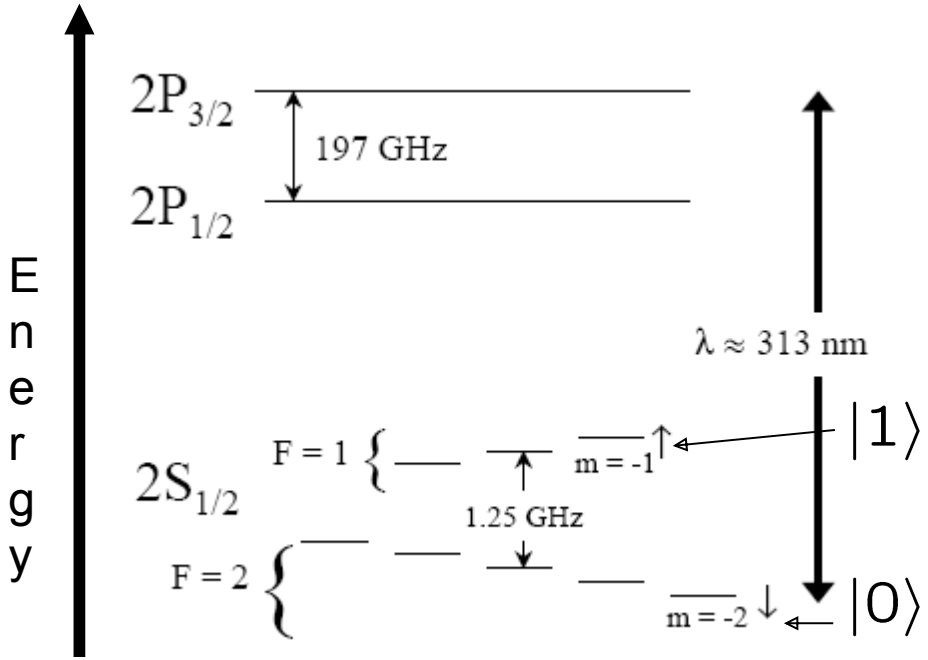


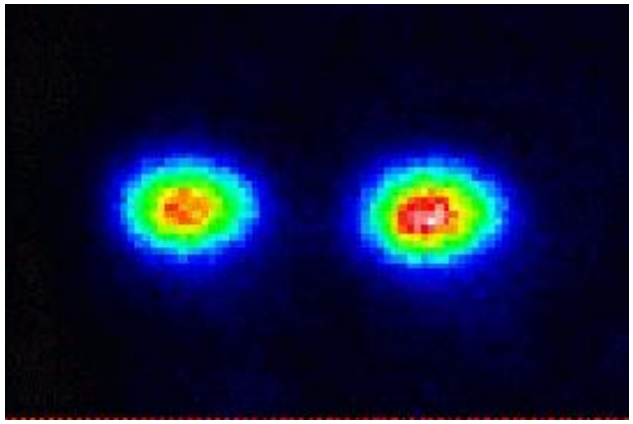


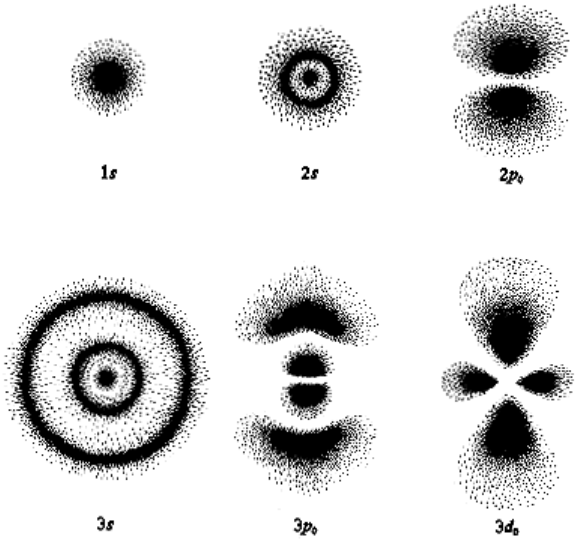
2 9Be+ Ions in an Ion Trap
(picture from NIST Boulder)



Where's the Qubit?





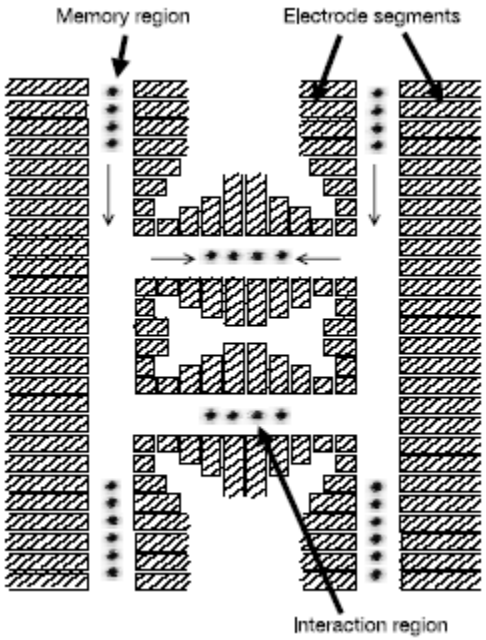


Each ion = 1 qubit

1. Well defined qubits ♠

orbitals

Scalable?



. Well defined qubits in a scalable architecture?

More to say about this later!!!

DiVincenzo's Criteria



David DiVincenzo

~~1. Well defined qubits in a "scalable" architecture~~

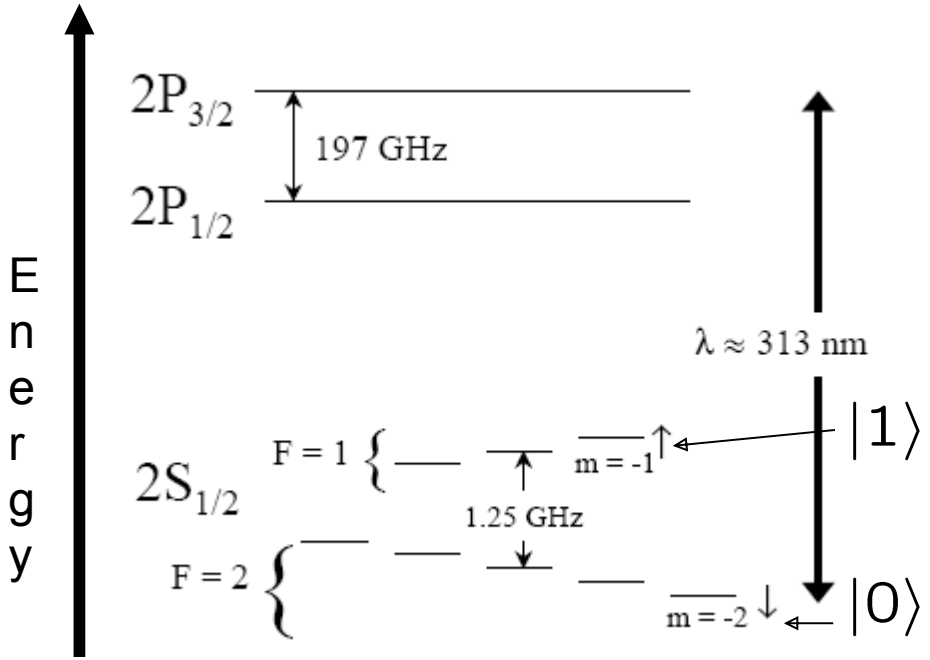
2. The ability to initialize the system to a state

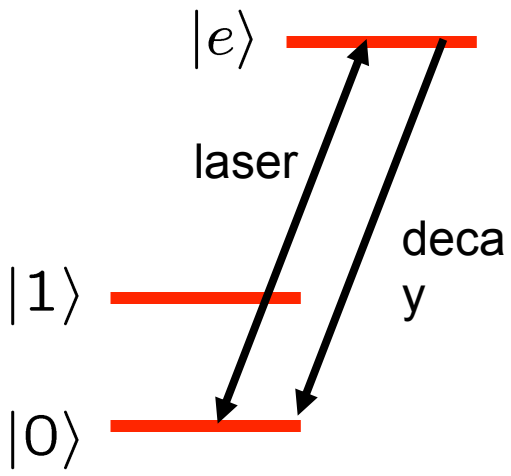
3. Have faster control over the system than error processes in the system.

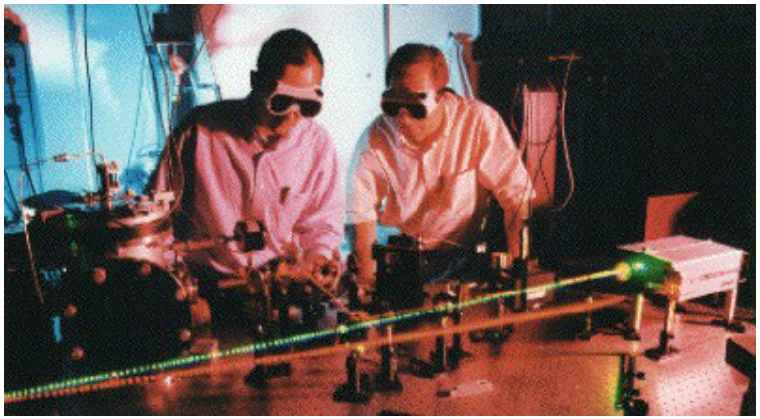
4. Have the ability to perform a universal set of quantum gates.

5. Have the ability to perform high quality measurements

Measuremen







Detecting florescence implies in state 0

99.99% efficiency

5. Have the ability to perform high quality measurements ♠

DiVincenzo's Criteria



David DiVincenzo

~~1. Well defined qubits in a "scalable" architecture~~

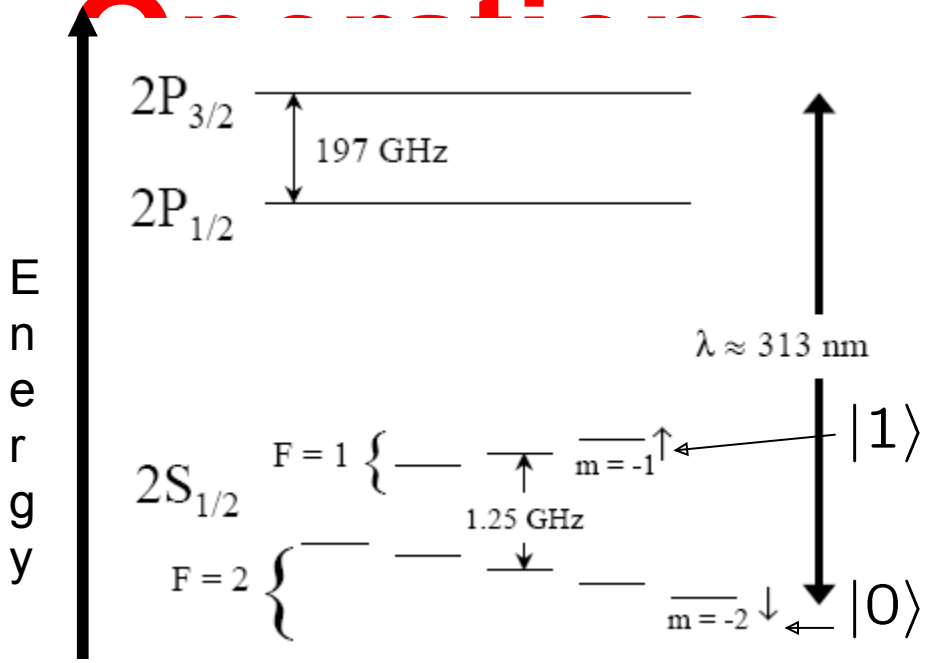
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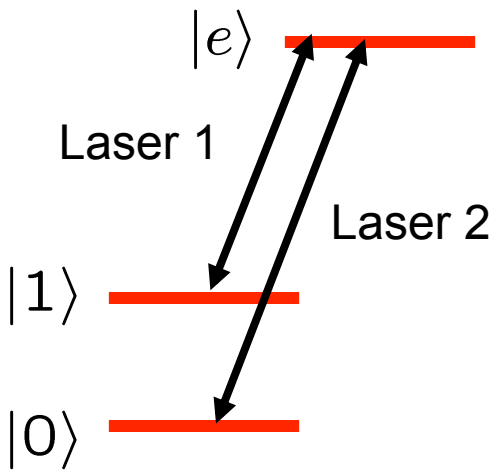
3. Have faster control over the system than error processes in the system.

4. Have the ability to perform a universal set of quantum gates.

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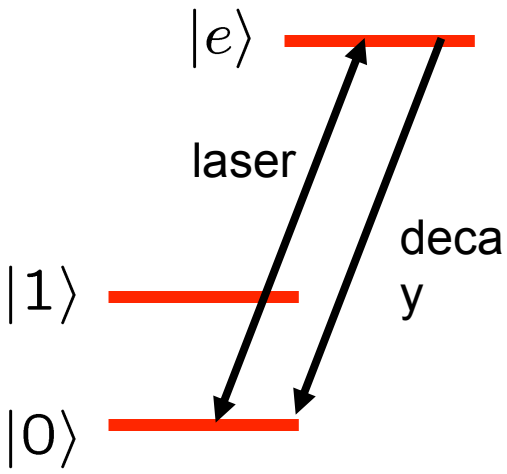
Single Qubit Operations



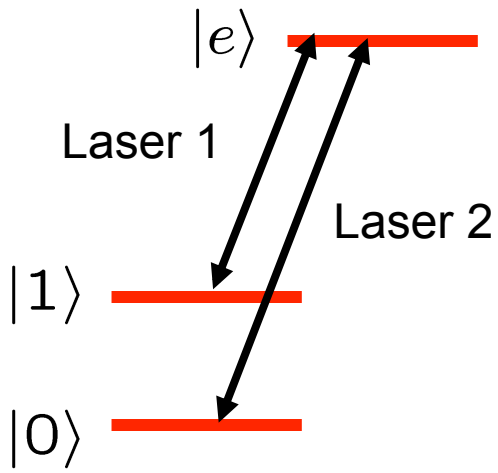


Allows any one qubit unitary operations $\begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}$

Initialization



measure



If not in zero state, flip

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

2. The ability to initial the system to a deterministic state. ♠

DiVincenzo's Criteria



David DiVincenzo

~~1. Well defined qubits in a "scalable" architecture~~

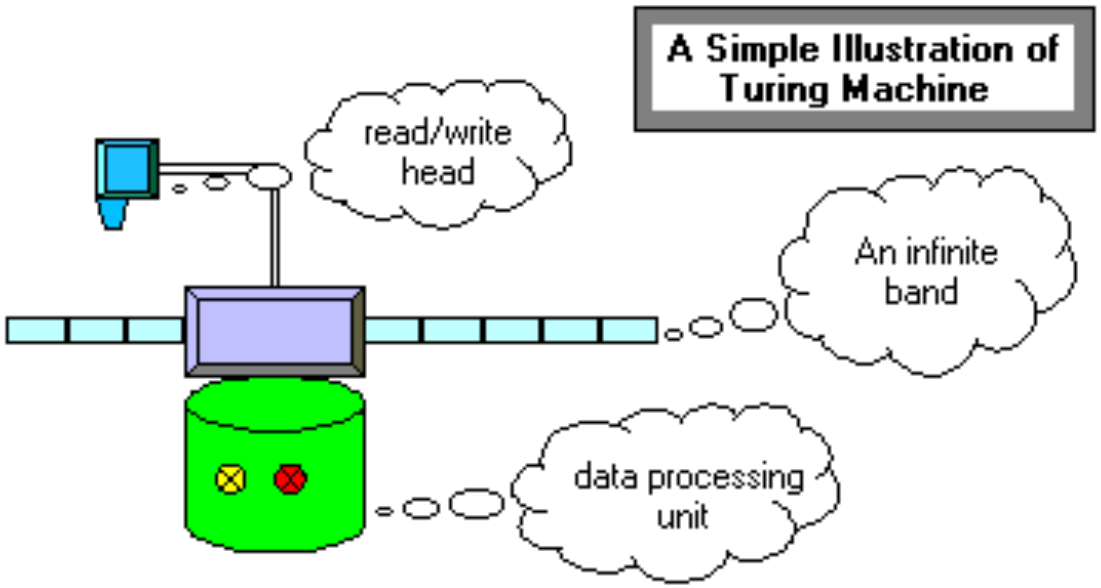
~~2. The ability to initialize the system to a state~~

3. Have faster control over the system than error processes in the system.

4. Have the ability to perform a universal set of quantum gates.

~~5. Have the ability to perform high quality measurements~~

Universal C



1. Turing machine reads state of tape at current position.
2. Based on this reading and state of machine, Turing machine writes new symbol at current position and possibly moves left or right.

A Universal Turing Machine can act like any other possible Turing machine (i.e. it is programmable)

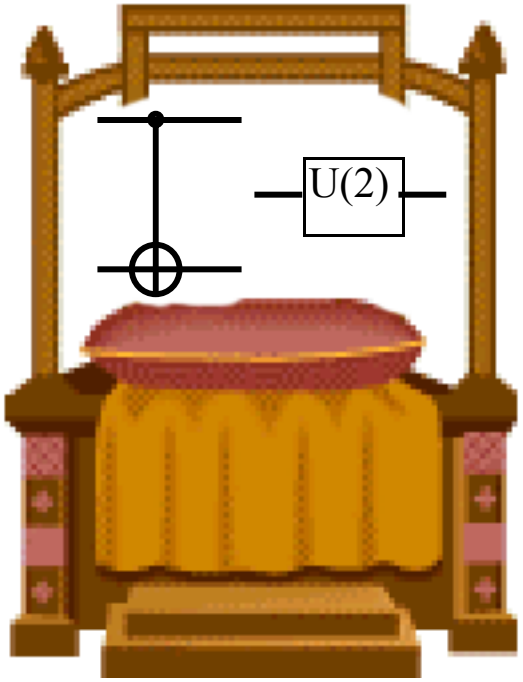
Universal Quantum Computer

Universal Quantum Computer

•a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

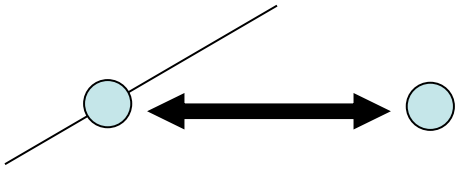
•a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.



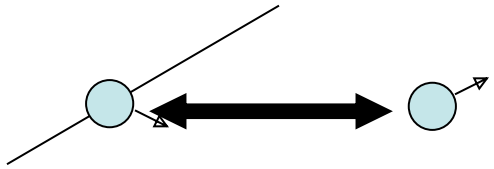
The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Coupling Two Qubits

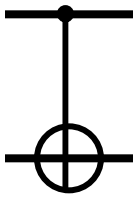


stationary



sloshing mode

These modes can be used as a bus between the qubits.



$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

4. Have the ability to perform a universal set of quantum gates ♠

DiVincenzo's Criteria



David DiVincenzo

~~1. Well defined qubits in a "scalable" architecture~~

~~2. The ability to initialize the system to a state~~

3. Have faster control over the system than error processes in the system.

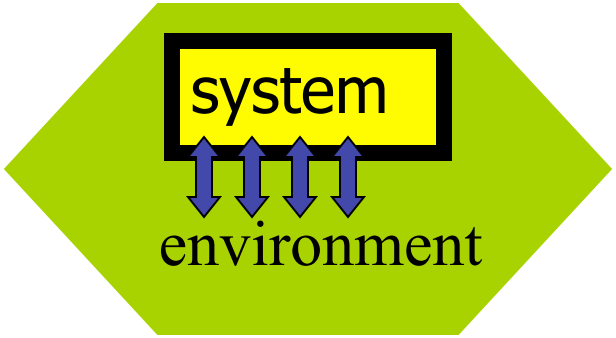
~~4. Have the ability to perform a universal set of quantum gates.~~

~~5. Have the ability to perform high quality measurements~~

What is the Problem?

3. Have faster control over the system than error processes in the system.

Real quantum systems are open quantum systems!



Quantum systems readily couple to an environment…

System decoheres:

$|0\rangle$
$|1\rangle$
qubits

$\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$

Quantum

0
1
pbits

50% **0** 50% **1**

Classical

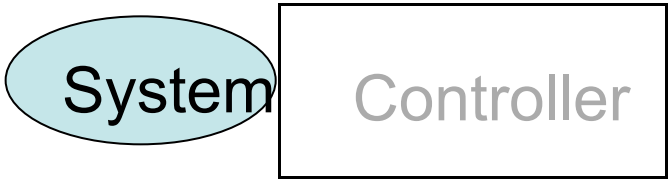
The Decoherence Problem

My Mortal Enemy

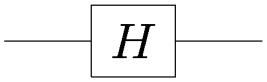


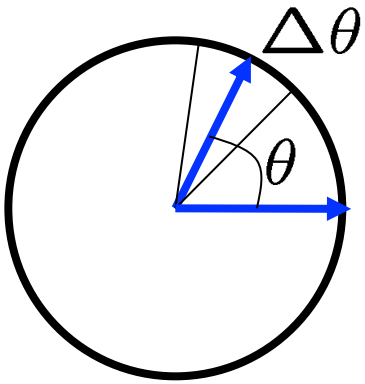
More Obstructions

FAULTY CONTROL



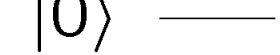
Imprecise or faulty unitary gates





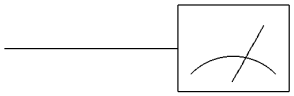
like not having control over probabilities…

Imperfect state preparation



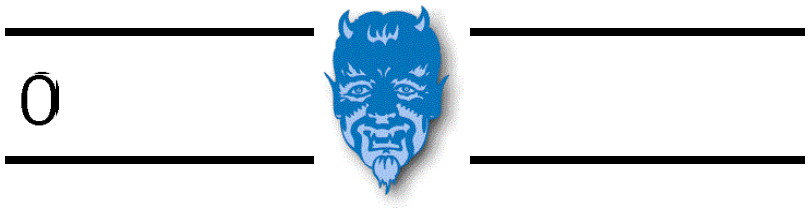
$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

Imperfect measurements



$$\alpha|0\rangle + \beta|1\rangle \rightarrow \begin{cases} Pr(0) = |\alpha|^2 - \epsilon \\ Pr(1) = |\beta|^2 + \epsilon \end{cases}$$

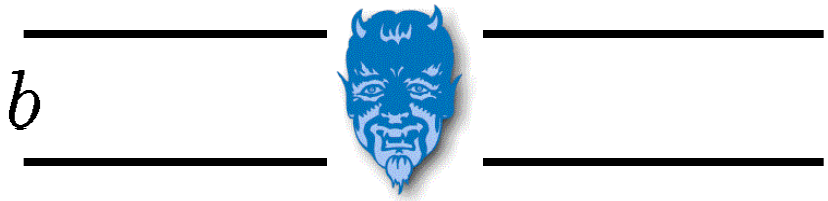
Classical Solutions



Probability of bit flip = *p*

Solution: redundancy

$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix}000\\001\\010\\100\end{matrix}\right\} \rightarrow 0$$

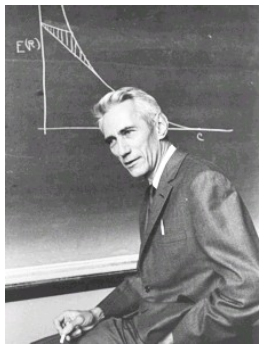
$$\left.\begin{matrix}111\\110\\101\\011\end{matrix}\right\} \rightarrow 1$$

Probability of failure =*3(1-p)p2+p3*

=3p2-2p3

<p if *p<1/2*

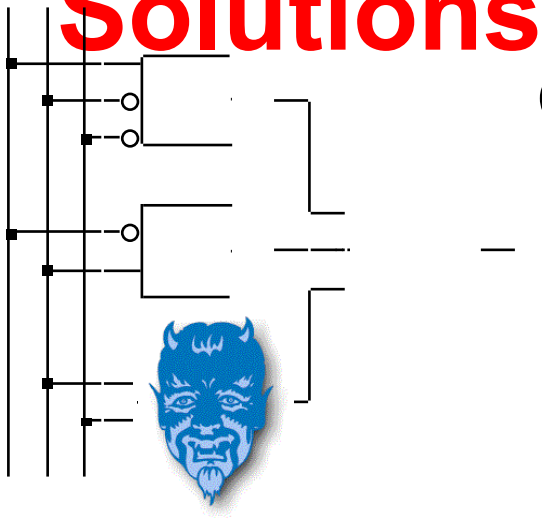
Use redundancy to protect information against noise



Claude Shannon 1948

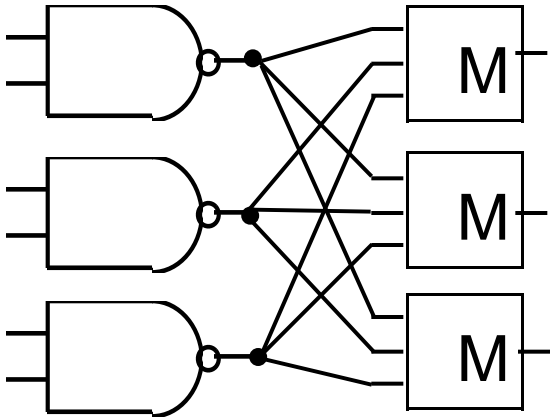
“A mathematical theory of communication”

More Classical Solutions



Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction





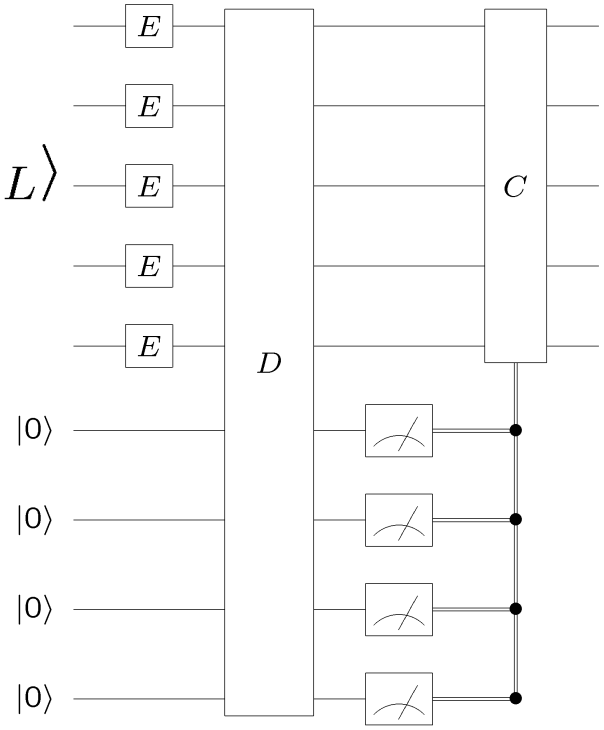
John von Neumann, 1956

“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

Quantum Error Correcting Codes

By encoding quantum information we can protect it from the effects of noise:

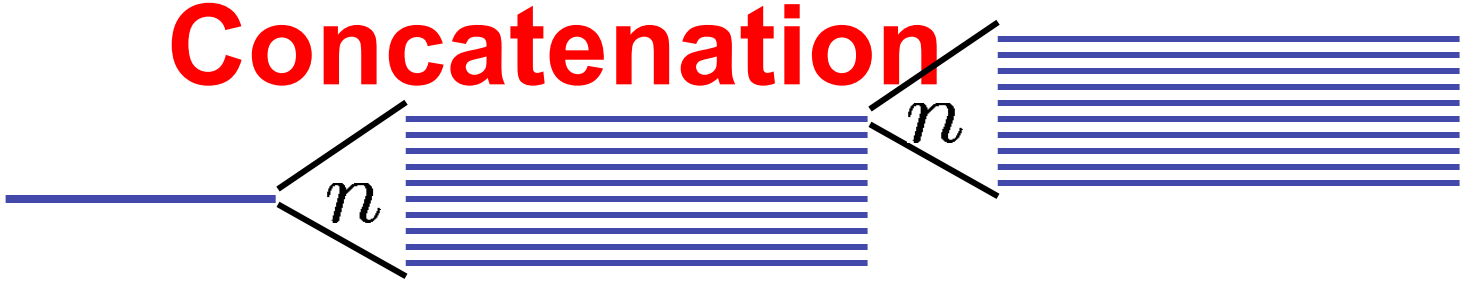
$\alpha|0_L\rangle + \beta|1_L\rangle$



$\approx \alpha|0_L\rangle + \beta|1_L\rangle$

"bare" error rate = *p*
error rate on encoded = *O(p2)*

Perfection Through Concatenation



failure probability

p $\quad$ $\frac{cp}{2}$ $\quad$ $\frac{c(cp2)}{2}$

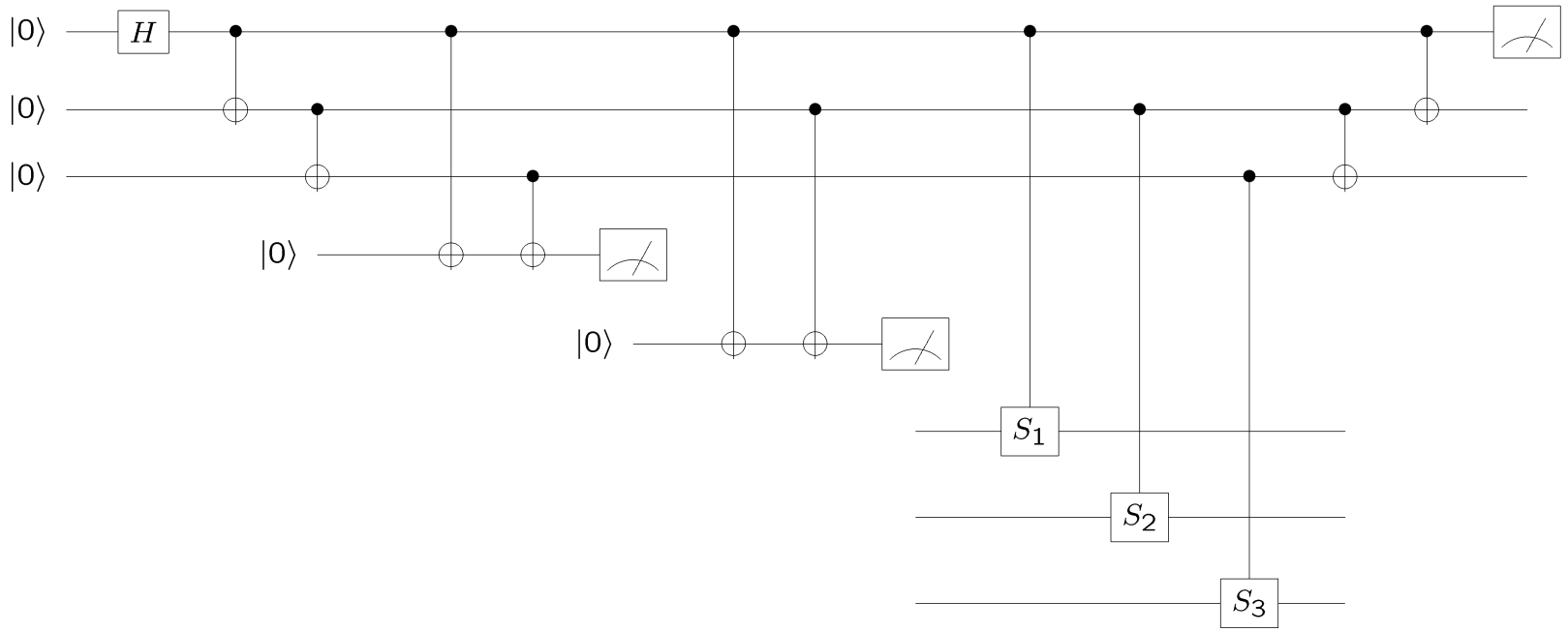
At kth level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}} \left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log \frac{1}{p_{fail}}\right)$$

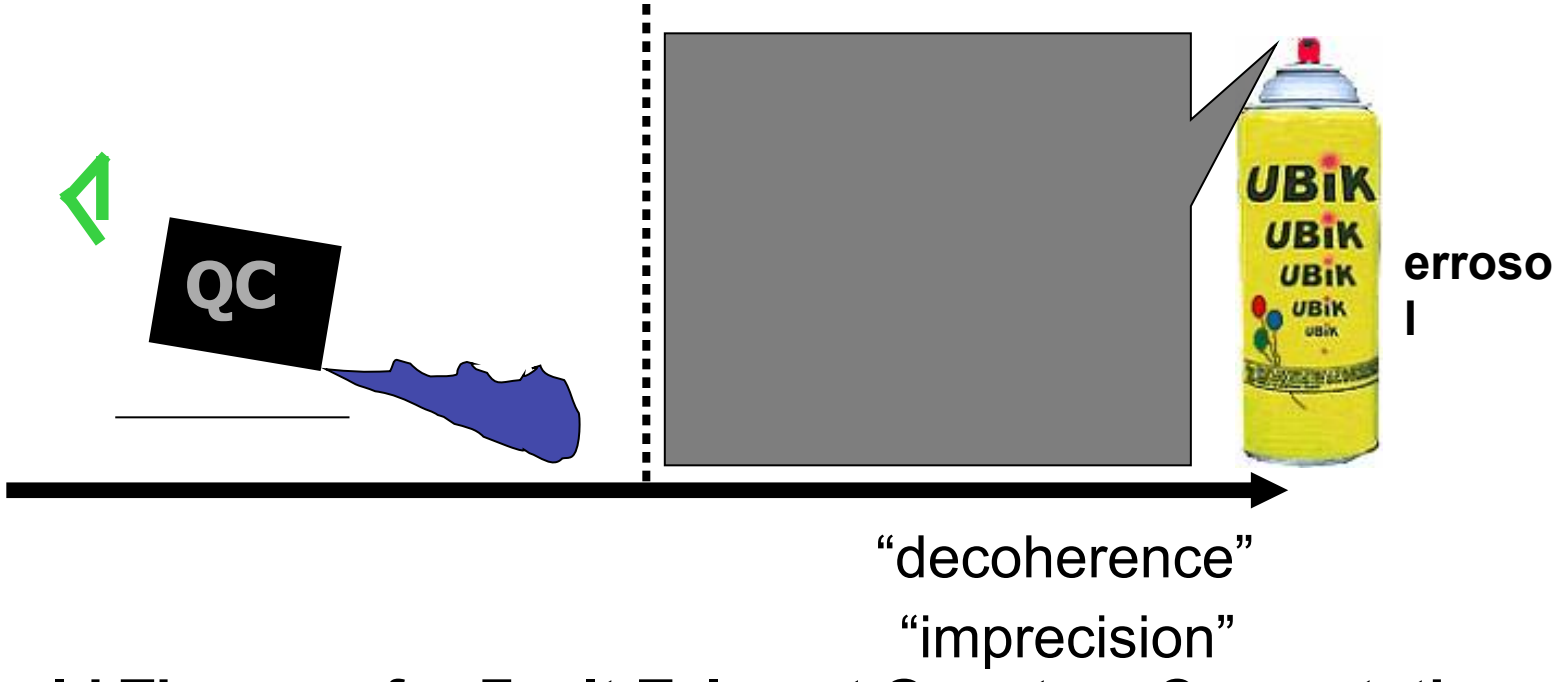
Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Threshold Theorem, ~1996



Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Threshold(s) Theorem(s)

Threshold theorems come in many different forms….

provable bounds, Monte Carlo estimates, …

different shapes…

assumptions of error models
assumptions of physical devices
assumptions of architectures

yielding threshold values for error rates from between 10-10 to 10-2

…but device complexities are universally very high. (hundreds to thousands to millions of “bare” qubits per encoded qubit with extremely complex manipulations)

Ion Trap Parameters

Operation	Time μs Now (Future)	Failure Rate Now (Future)
Single Gate	1 (1)	10-4 (10-8)
Double Gate	10 (10)	0.03 (10-7)
Measurement	200 (10)	0.01 (10-8)
Movement	20 (10)	0.005 (10-7)/ μm
Split	200 (0.1)	
Cooling	200 (0.1)	
Memory	10 to 100 s	
Trap Size	~200 (1-5) μm	

from Thaker et. al. ISCA-33

DiVincenzo's Criteria



David DiVincenzo

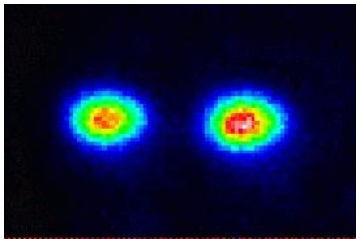
1. Well defined qubits in a "scalable" architecture

2. The ability to initialize the system to a state

3. Have faster control over the system than error processes in the system.

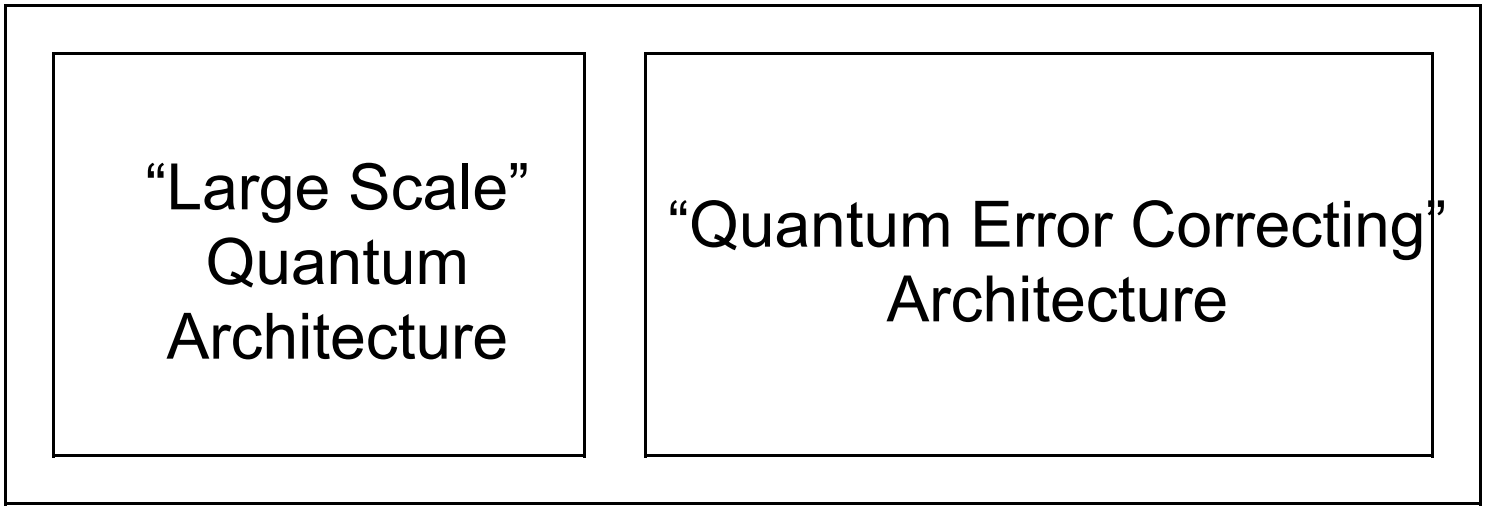
4. Have the ability to perform a universal set of quantum gates.

5. Have the ability to perform high quality measurements



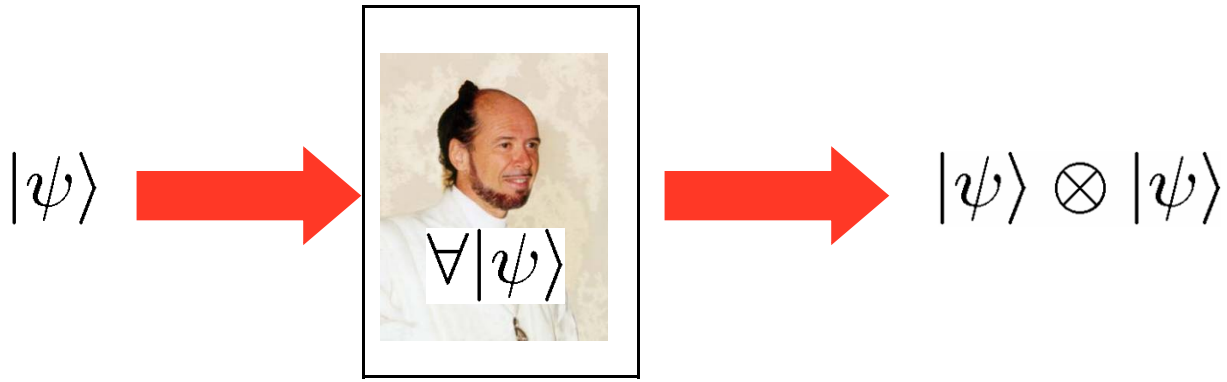
Ion traps "satisfy" all requirements but…

Quantum Computing Architectures



A Major Issue

Qubits cannot be copied:



No cloning theorem

Qubits must be moved from place to place, interact with other qubits, etc.

Moving Qubits

SWAP channel?



Problem: bandwidth decays exponentially with distance

(SWAP error multiply)

Move the qubits?



Problem: qubits heat up when you move them

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

1 qubit

X

Y

2 bits





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ ○ $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ ○ $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Alice performs operation on her state and half of entangled system and sends outcome

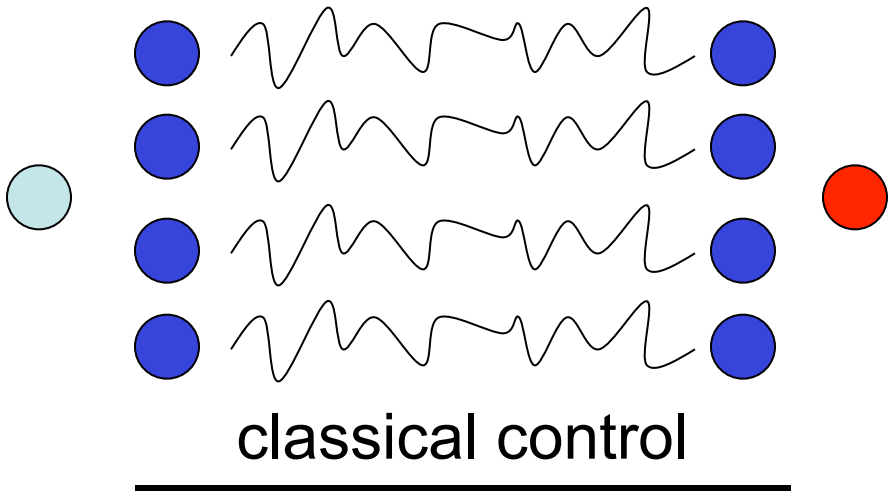
Conditional on received bits, Bob performs manipulation of his half of entangled system

Moving Qubits

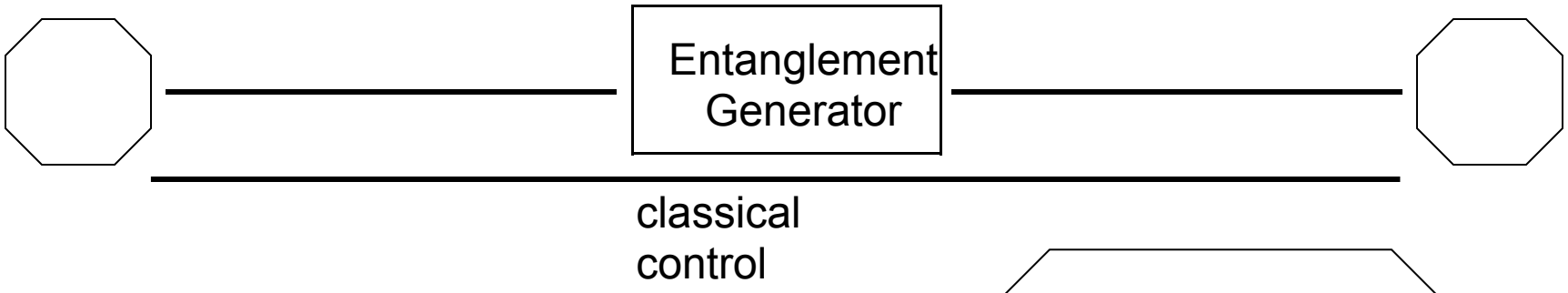
Teleportation channel



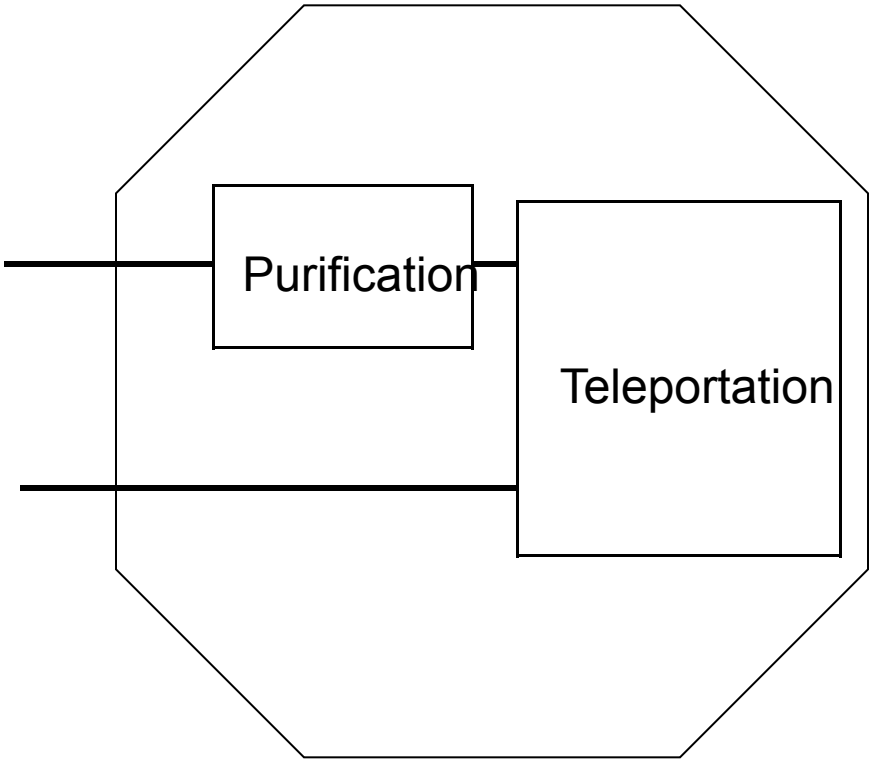
Entanglement distillation



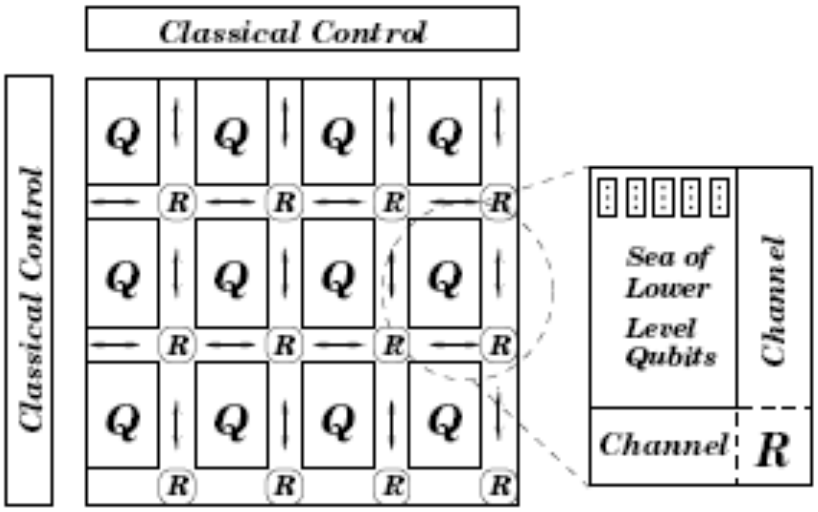
Teleportation Units



For large scale architecture teleportation provides a robust method for moving quantum information



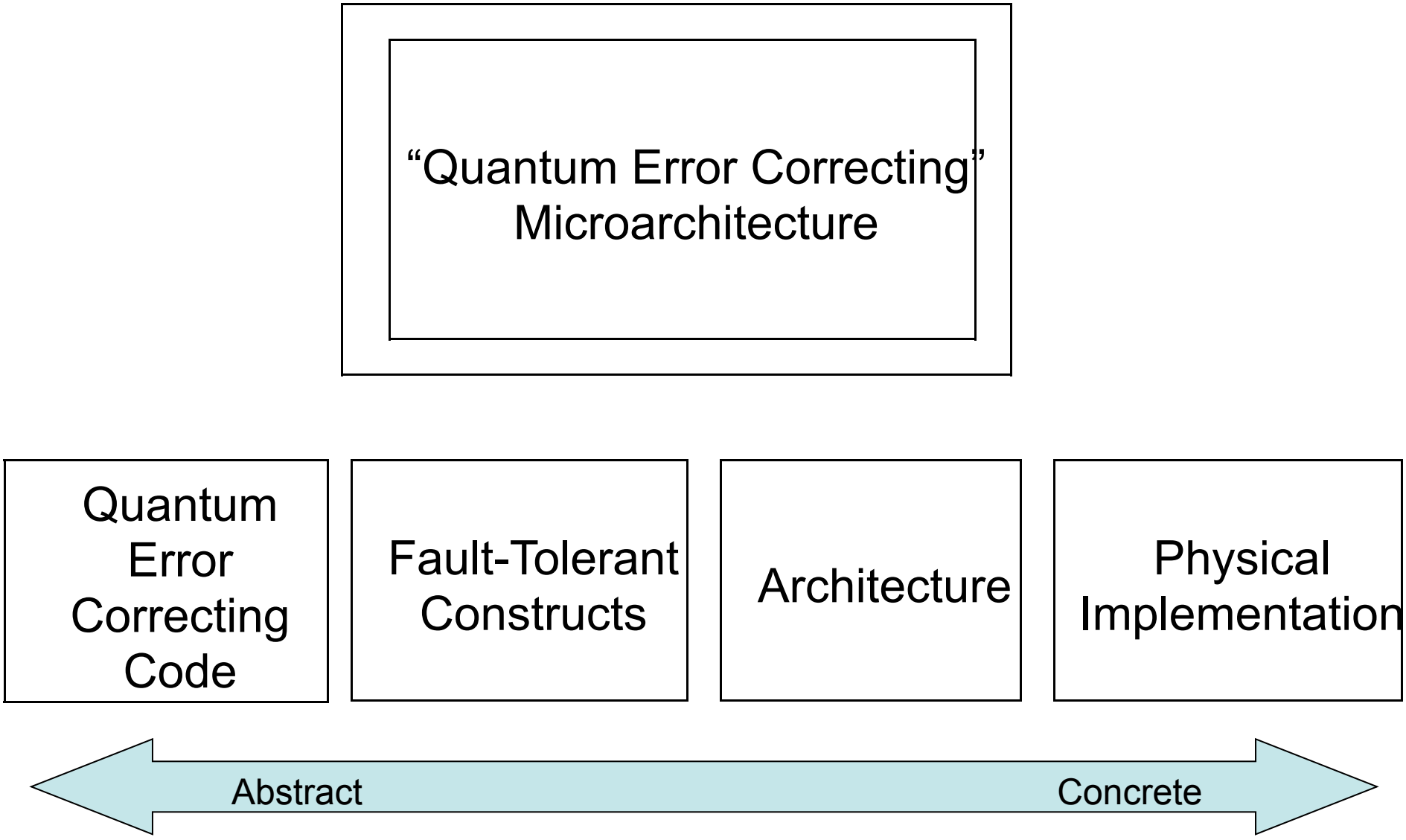
Quantum Logic Ar



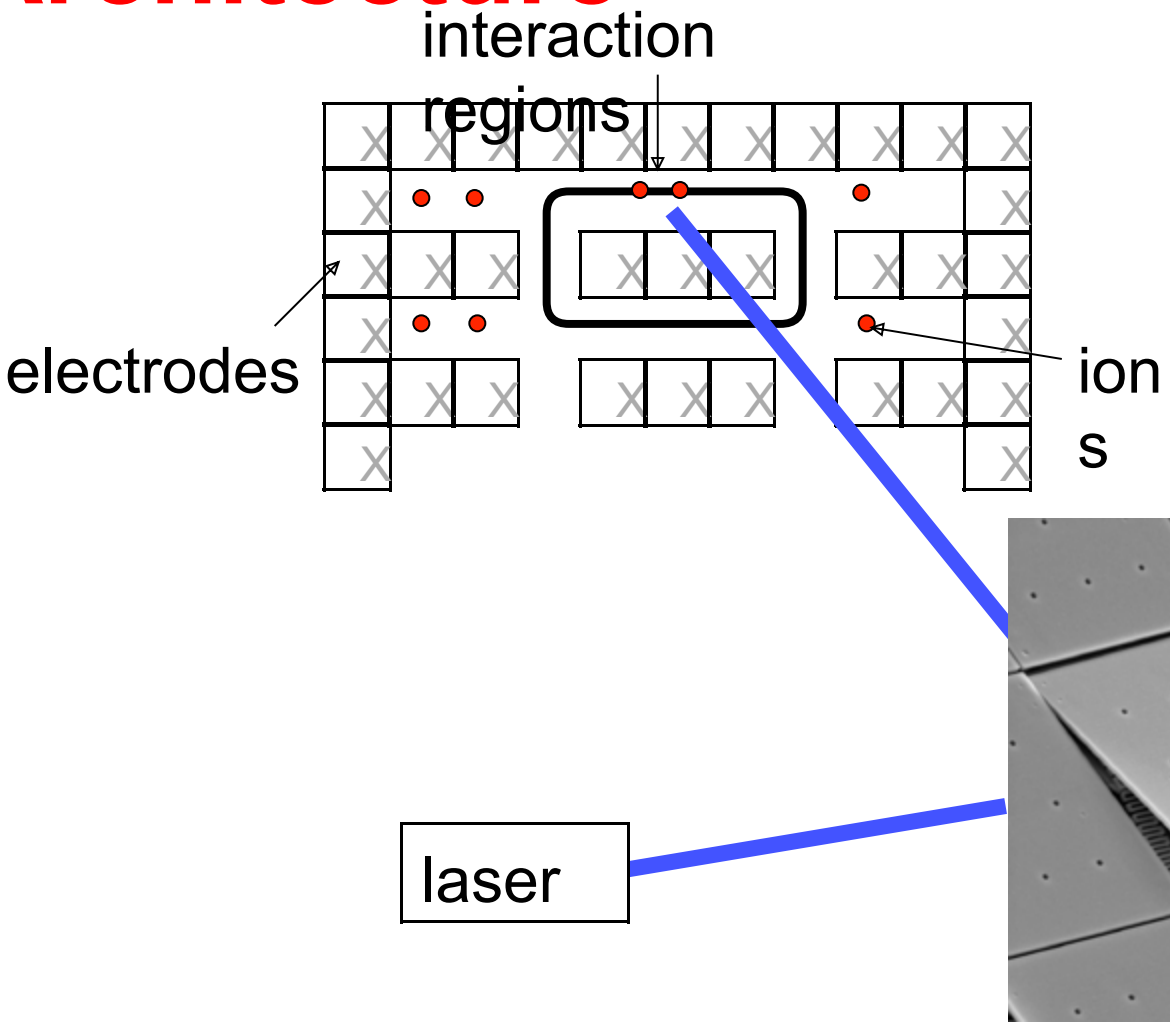
(from Metodi et al, MICRO-38)

Build quantum computer from array of uniform blocks. These blocks can be configured as teleportation units, quantum gates, preparation, and measurement. But each element will have a serious quantum error correcting microarchitecture.

Correcting, Correcting, Correcting



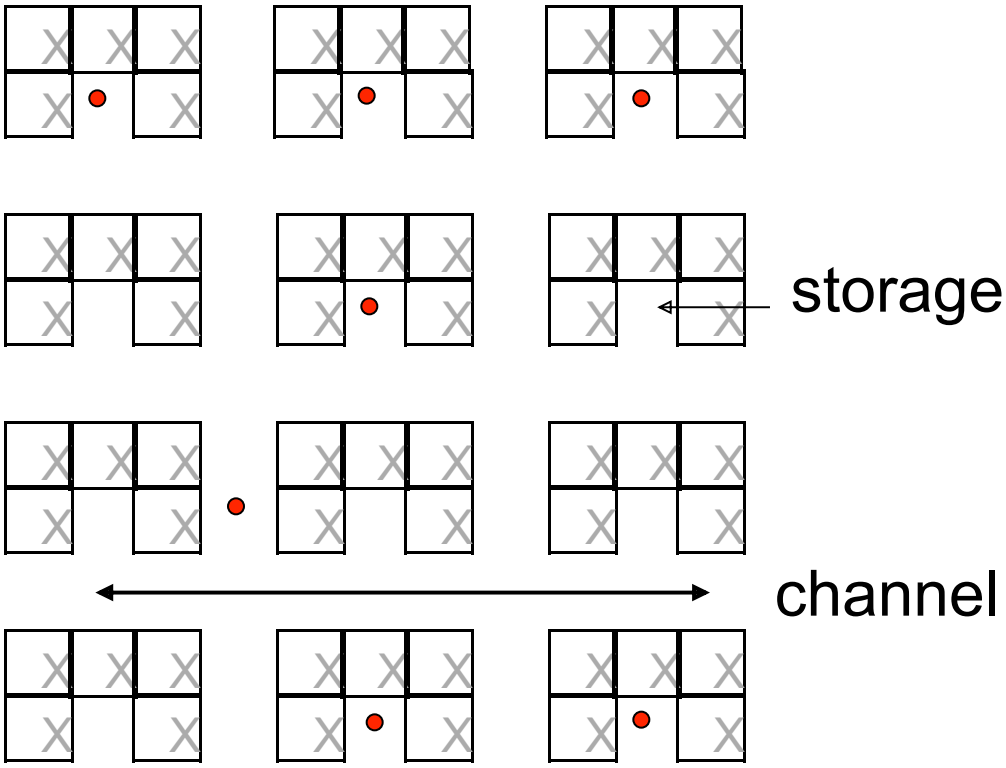
Quantum CCD Architecture



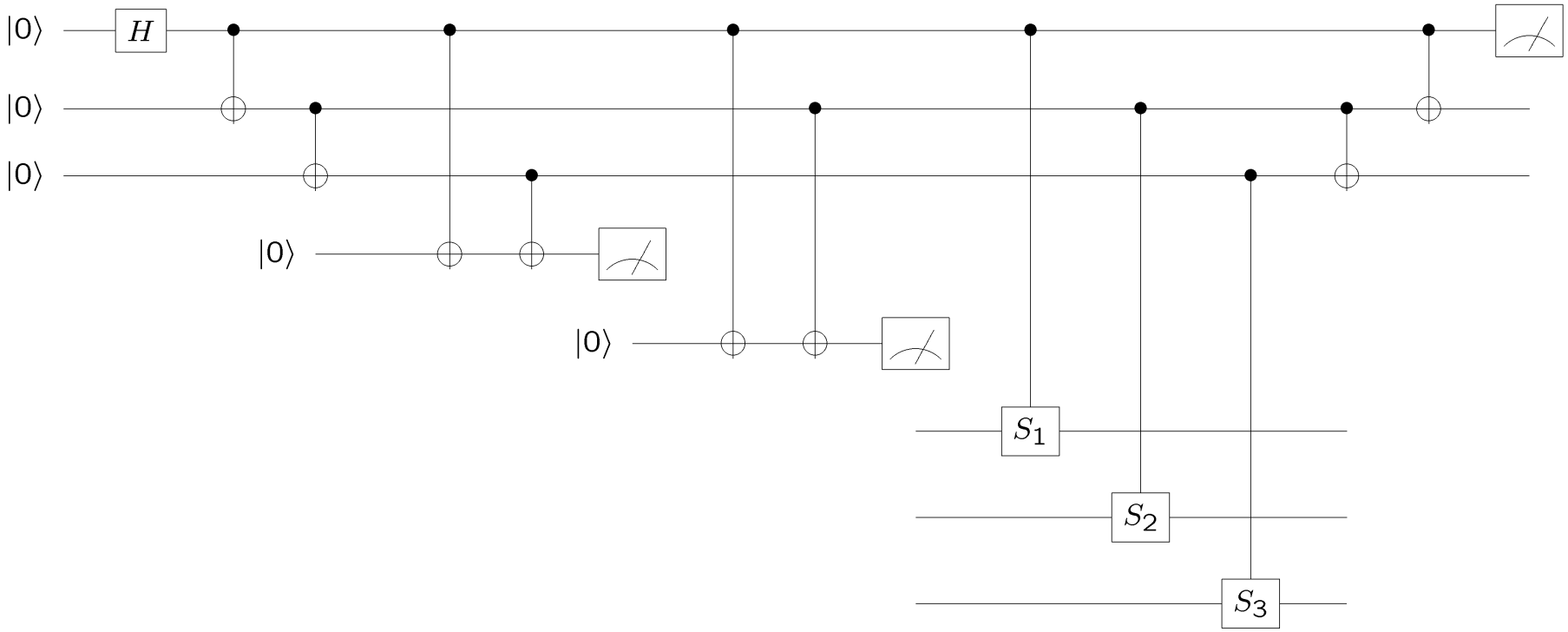
Kielpinski *et al Nature* 2002

MEMS mirror control (example from Sandia)

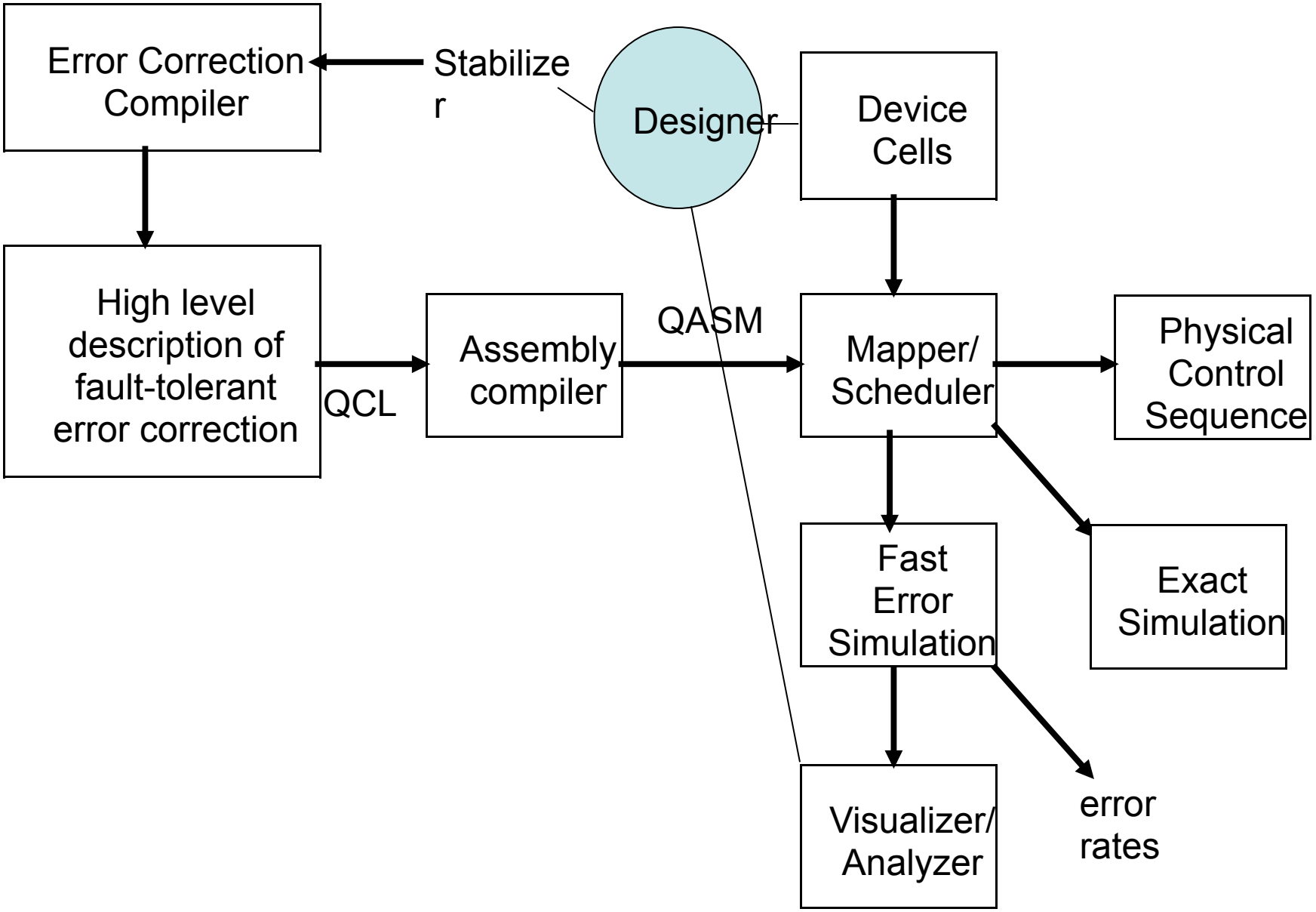
Simple Block Architecture



Fault-tolerant Procedures



Tools!


Error Correction Compiler
Stabilizer
Designer
Device Cells
High level description of fault-tolerant error correction
QCL
Assembly compiler
QASM
Mapper/ Scheduler
Physical Control Sequence
Fast Error Simulation
Exact Simulation
Visualizer/ Analyzer
error rates

High Level Languages

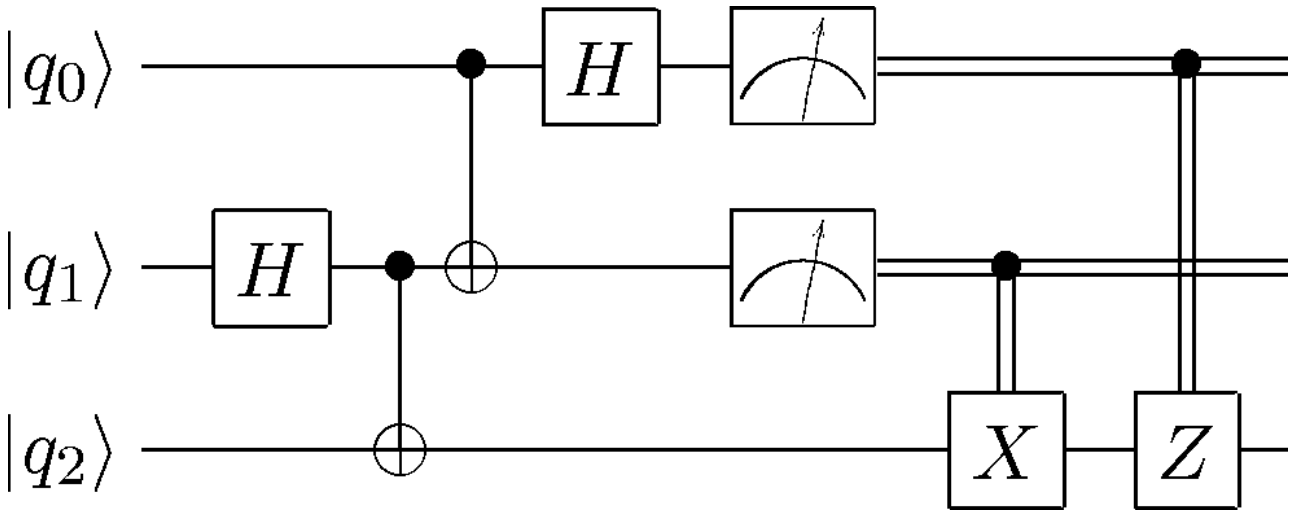
Example in QCL (http://tph.tuwien.ac.at/~oemer/qcl.html)

```
operator dft(qureg q) { // main operator
const n=#q;                  // set n to length of input
int i; int j;              // declare loop counters
for i=0 to n-1 {
   for j=0 to i-1 {          // apply conditional phase gates
      CPhase(2*pi/2^(i-j+1),q[n-i-1] & q[n-j-1]);
   }
   Mix(q[n-i-1]);            // qubit rotation
}
flip(q);                        // swap bit order of the output
}
```

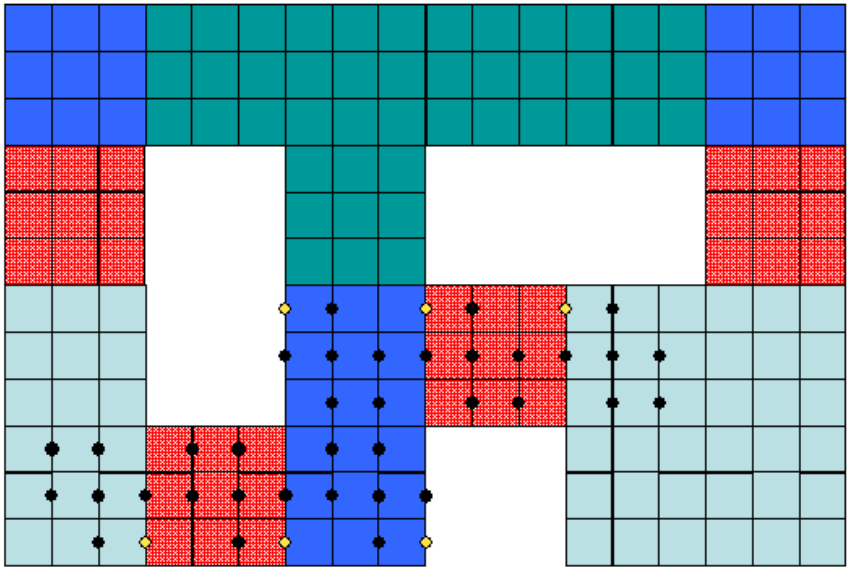
Quantum Assembly Language

```
qubit q0
qubit q1
qubit q2
h q1        # create EPR pair
cnot q1,q2
cnot q0,q1    # Bell basis measurement
h q0
nop q1
measure q0
measure q1
c-x q1,q2    # correction step
c-z q0,q2
```

QASM



Schedule
r

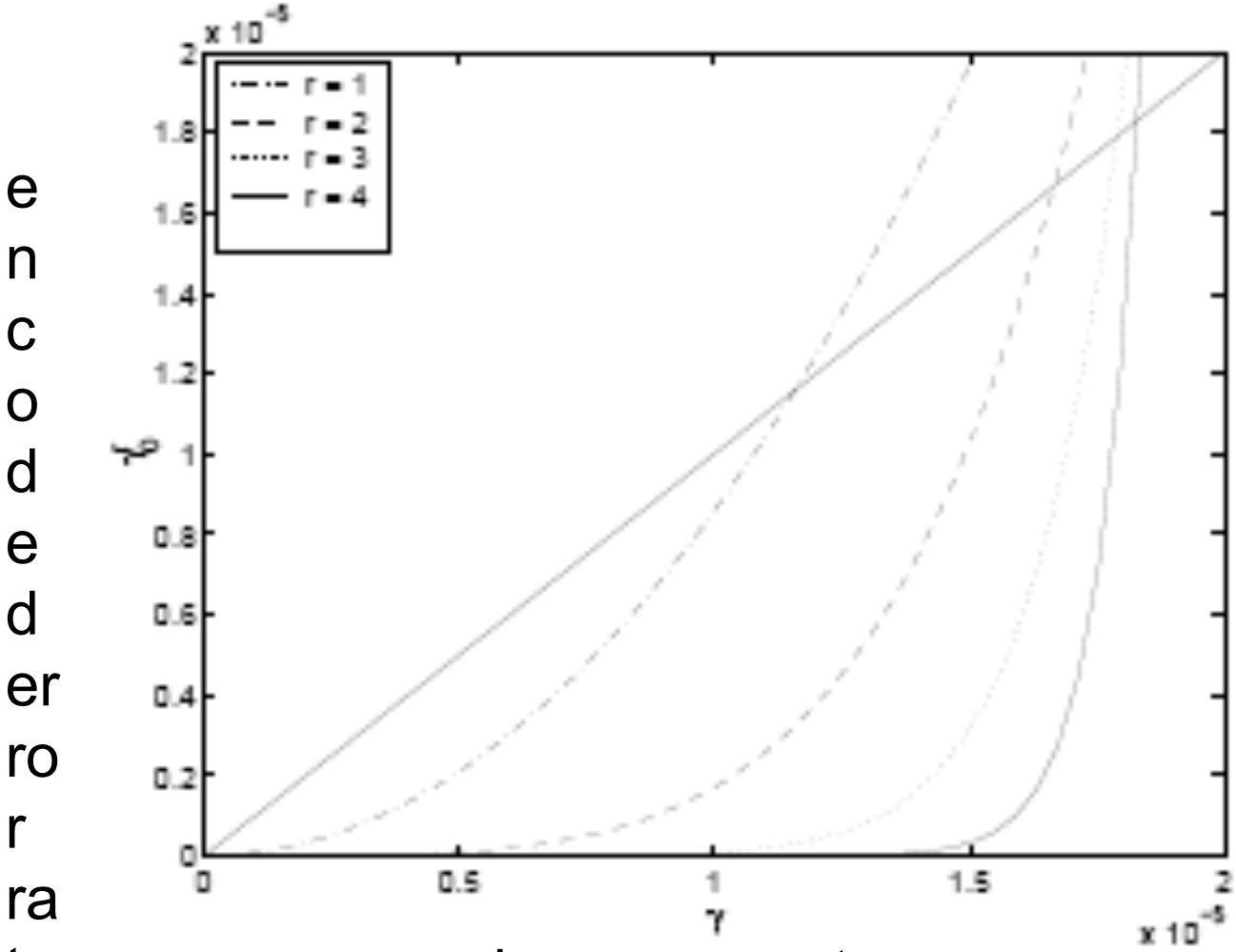


Z/X 1-EC

Example from Chuang group at http://www.media.mit.edu/quanta

Metrics

Main goal for QEC microarchitecture is to extract error rate



encoded error rate

bare error rate

from Svore *et al* 2006

Brave New World

Designing entire architecture from QEC to large scale task

Input Size	Compute Blocks	Area Reduced (Factor of)		SpeedUp		Gain Product	
		St-Code	BSr-Code	St-Code	BSr-Code	St-Code	BSr-Code
32-bit	4	6.69	9.80	0.54	1.47	3.61	14.41
	9	3.22	4.74	0.97	2.9	3.14	13.74
64-bit	9	6.36	9.32	0.70	1.92	4.45	17.70
	16	3.79	5.56	0.98	3.0	3.71	16.68
128-bit	16	7.24	10.6	0.72	1.97	5.24	20.88
	25	4.90	7.17	0.96	2.84	4.70	20.36
256-bit	36	6.65	9.47	0.92	2.51	6.12	23.68
	49	5.07	7.43	0.98	2.98	4.96	22.14
512-bit	64	7.42	10.87	0.92	2.50	6.80	27.18
	81	6.06	8.87	0.98	2.91	5.94	25.81
1024-bit	**100**	**9.14**	**13.4**	**0.80**	**2.19**	**7.35**	**29.35**
	121	7.81	11.45	0.97	2.65	7.60	30.34

Table 4. For various size inputs, this table shows how the CQLA performs for Modular Exponentiation. The space saved due to compressing the memory blocks and separating memory and compute regions is shown as compared to prior work [1]. St-Code is the Steane ECC and BSr-Code is the Bacon-Shor code. The Gain Product is compared with our prior work, the QLA, which has a Gain Product of 1.0.

Thaker *et al* ICSA 33

Major Challenges

•Quantum error correction microarchitecture

- Design of quantum error correcting codes
- Design of fault-tolerant procedures (lack systematic theory)

- Designing cells customized to error correction
- Designing cells to maximize parallelism in error correction

- Designing quantum error correction customized to physical parameters

- Macro quantum architecture (largely unexplored)

- Maximization of parallelism at circuit level
- Everything you've learned in this class!

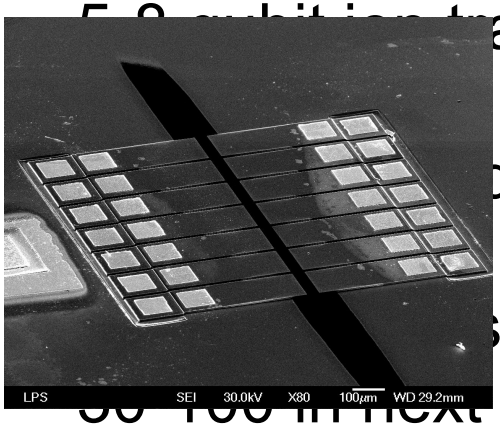
Quantum Architectures



Today: a taste

On the verge of becoming extremely important

- 5-8 qubit ion trap machines
- Scaled ion traps
- Will scale up to traps with 50-100 in next few year



Picture from

Architects to the Rescue!





Land of
Large
Quantum
Computers



More than a
few qubits